

## Soft

IEEE defines the software engineering as - The application of a systematic, disciplined, quantifiable approach to the development, operation and maintenance of software i.e. application of engineering to software.

Software engineering is the systematic approach to the development, operation maintenance and retirement of the software.

Another definition given by ~~BOHEH~~ BOEHM 's - software engineering is the application of science and mathematics by which the capabilities of computer equipment made useful to man by a computer programs, procedures and associated documentation.

SE can also define as a procedure whose aim is the production of quality software, software that is delivered on time within budget and that satisfy its requirement.

The primary object of S.E. is to produce good quality software with

minimum cost within the limited  
also time.

There is a difference between S.E and programming. Programming is the part of S.E. If we write a program it cannot be S.E. because S.E. requires systematic and cost effective approach for developing good quality program.

Software Process - The term process means a particular method of doing something generally involving a no. of steps or operations. In software process refers to the method of developing software.

A software process is a set of activity together with ordering constraints among them, such that if the activities are performed properly and in accordance with the ordering constraints the desired result is produce.

In an organisation there are typically many processes immed simultaneously executing many of these do not concern software engineering. This could may considered non-software engineering process

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model.

The process that deals with the technical and management issue of software development is called.

Clearly, many different type of activities need to perform to develop software. All these activities together ~~summ~~ summarise software process.

The three generic phases of software engineering include -

1. The definition phase
2. The Development Phase
3. The ~~Maintain~~ Maintenance Phase

1. The Definition phase focuses on what. that is during definition the software developer attempts to identify what processed and what function and performance is desired what system behaviour can be expected what interface are to established, what design constraints, what validation criteria required to define a successful system.

This phase include 3 main tasks -

- (i) → System or information engineering.

- (ii) → Software Project Planning.  
(iii) → Requirement analysis.

The Development phase focuses on how that is during the software development a software engineering attempts to define how data are structured how function is implemented

how procedure are to be implemented how design will be translate into a programming language, how testing will be performed. This phase includes 3 main tasks -

- 1) Software design
- 2) Code generation
- 3) Software testing

Maintenance - The maintenance focuses on change that is associated with the software. The 4 main change that are encountered during this phase are -

- 1) Correction - It is rightly that the customer will find errors or defects in the software inspite of the quality assurance activities corrective maintenance changes the software to correct defects.

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2 → Adaptations - As time progresses it is likely that the original environment (for ex. CPU, Operating system, business rules etc) for which the software was developed is likely to change. Adaptive maintenance results in modification to the software to ~~accom~~ accommodate changes to its external environment.

3 → Enhancement - As software is used, the customer will recognise the additional functional requirements that will benefit him. This type of maintenance extends the software beyond its original functional requirements.

4 → Prevention - Computer software deteriorates due to changes. So preventative maintenance often called software re-engineering must be conducted in order to make changes to the computer software more easy.

Software Crisis :- The software crisis was a term used in the early days of software engineering, before it was a well established subject. It refers to the difficulty of writing correct, understandable and verifiable computer program. The roots of the

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software crisis are complexity, expectation and changes.

The causes of the software crisis are pre-linked to the overall complexity of the software causes and the relative immaturity of software engineering as a profession. The crisis generated in similar ways -

- 1 → Projects running overbudget
- 2 → Projects running overtime.
- 3 → Software often did not meet the requirement.
- 4 → Software was of low quality.
- 5 → Projects were unmanageable and code difficult to maintain.

Various causes and methodologies have been developed over the last few decades to control the software crisis with varying degree of success.

SDLC (Software Development Life Cycle) -

A software life cycle is a series of identifiable stages that is software product under goes during its lifetime. Each of these stage is called life cycle phase.

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SD SDLC consist of following phases during its life cycle -

- 1) Requirement Analysis
- 2) Design
- 3) Coding
- 4) Testing
- 5) Implementation
- 6) Maintenance

1) Requirement Analysis - Identification of the requirements for the system is the first stage of SDLC. This activity consist of first gathering the requirements and then analysing the gathered information. The goal of the requirements gathering activity is to collect all relevant information regarding the product to be developed.

2) Design - It is the most creative and challenging face of SDLC - the term design describe a final logical system. Design phase of software development deals with transforming the customer requirement in a logical system we can classify various design activities into 2 important parts

(i) System Design - A system design which is sometime also called Top Level Design aim to identify

the module that should be in the system, specification of this model and how they interact with each other to produce the desired result.

(ii) ⇒ Detailed Design - During this phase further details of the data structure and algorithmic design of each of the module is specified.

In system design attention is on what components are needed, why is detail design, how components can be implement in the software is main issue.

Coding - Once the design is completed the most of the major decision about the system have been made the goal of the coding phase is to translate design of the design into code in given programming language coding phase affect both testing and maintenance.

Well written can reduce the testing and maintenance effort. during coding the focus should be on developing programs that are easy to read and understandable.

Im Testing - Testing is the major quality control during the software development. Its basic function is to detect the errors from the software. The role of testing is to uncover requirement, design and coding errors in the program.

Types of Testing - There are several types of

i) Unit Testing - starting point of the testing is unit testing each module is tested separately perform by the coder himself simultaneously with the coding of the module.

Integrating Testing - This testing is perform to detect design errors by focusing on testing the interconnection b/w analysis and design.

System Testing - After a system is put together system testing is perform. System is tested against the system requirement to see if all the requirement are match. sys. testing usually consist of diff 3 different type of activities.

Alpha testing - is the system testing performed by development team. (c)

Beta testing - This is the system testing performed by a friendly set of customers.

Acceptance Testing - This is the system testing performed by the customer himself after the product delivery to determine whether to accept the delivery of product or reject it. Acceptance testing is performed to demonstrate to the client on the real life data and operation of the system.

Implementation - The implementation phase is less creative than system design. It is primary concern with user training, site preparation and file conversion. There are 3 type of implementation -

- (a) Implementation of the computer sys. to replace a manual system.
- (b) Implementation of new system to replace an existing one.

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(c)  $\Rightarrow$  Implementation of modified application to replace an existing one using the same computer.

Maintenance - Maintenance of a typical software product requires much more efforts than the efforts necessary to develop the product itself. Many studies carried out in past confirm this and indicate that the relative effort of development of a software product to its maintenance efforts is roughly in 40:60 ratio. Maintenance involves following kinds of activity -

Corrective - This type of maintenance involves correcting errors that were not discovered during the product development life cycle phase.

Perfective " :- This type of maintenance involves improving the implementation of the system and enhancing functionality according to the customer requirements.

Adaptive " :- Porting to a new computer or a new operating system is called adaptive maintenance.

Adaptive main. is usually required for porting the software to work in a new environment.

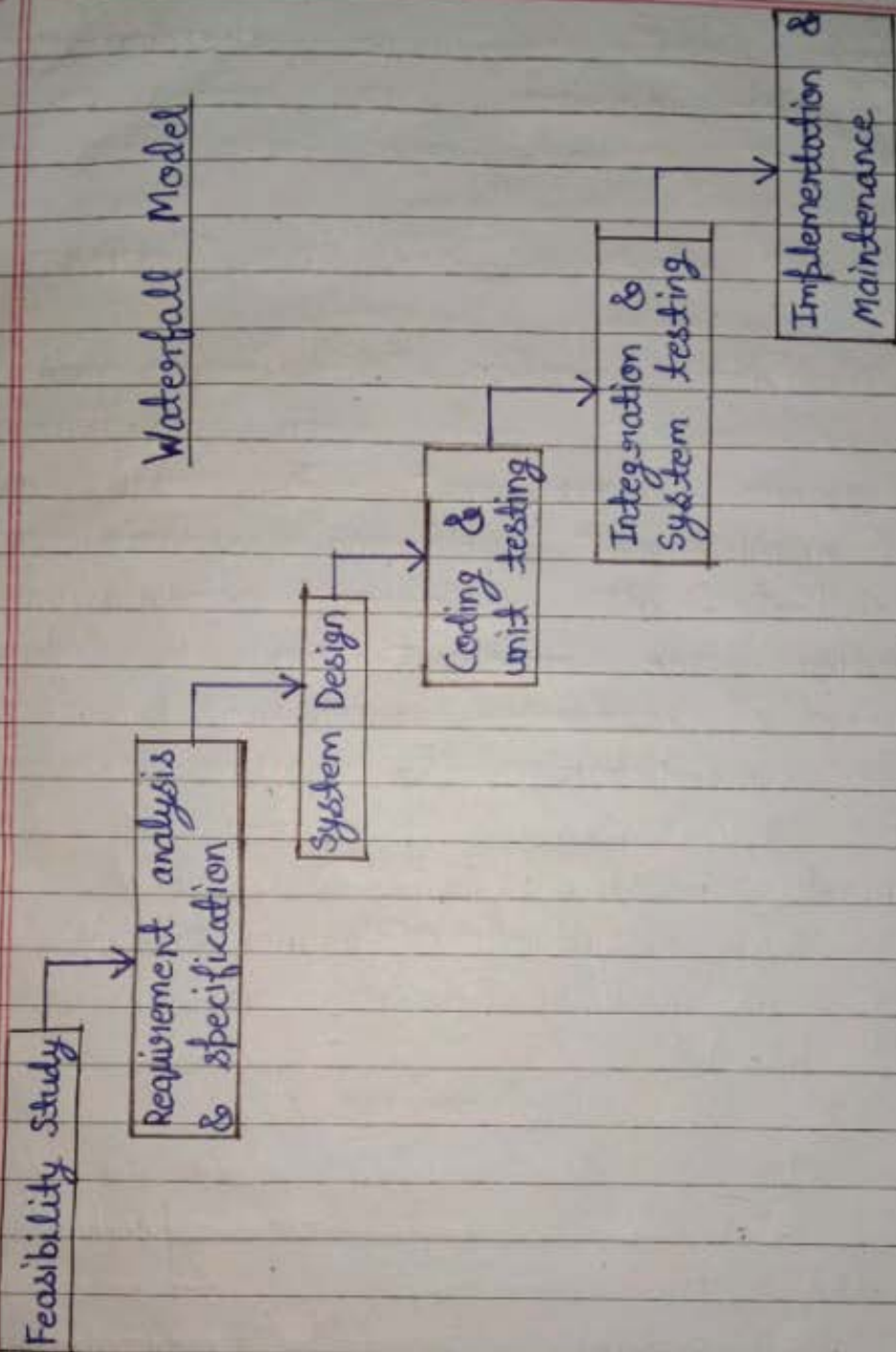
## Software Development Process Model/ Software Life cycle model -

The softwares are essentially a step of steps such as process, methods and tools and these steps are referred to as software engineering paradigm or software life cycle model. The few software life cycle model are are -

- i) Waterfall Model
- ii) Prototype Model
- iii) Evolutionary Model
- iv) Spiral Model

Waterfall Model - The waterfall model is a sequential software development model. It is not a practical model in the sense that it can not be used in actual software development project. That is, we consider this model to be a theoretical way of developing software. All other life cycle models in some way are based on the classical waterfall model. The different

## Waterfall Model



levels of the waterfall model correspond to the distinct phase of the software development process.

The phases of waterfall model -

The main aim of the feasibility study

activity is to determine whether it would be financial and technical feasible to develop the product.

### Requirement Analysis and Specification -

This activity consist of first gathering the requirement and when analysing the gather requirement. The goal of the requirement gathering activity is to collect all relevant information regarding the product to be developed from the customer with a view to clearly understand the customer requirement. This phase produces a large document written in a natural language. The resultant document is known as SRS (Software Requirement Specification).

The goal of the design face is to transform the requirements specified in the SRS document in to structure that is suitable for implementation in some programming language. there are 2 type of design present at time -

#### i) Traditional Design Approach -

The traditional design approach is currently being used by many software developer houses

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The traditional design technique is based on the data flow orientated design approach.

- ii) Object Orientated :- object orientated design is a relatively new technique various objects that occur in the problem domain are first identified and the different relationships that exist among these objects are identified.

### Coding and Unit Testing -

During this phase coding proceeds smoothly because all the information needed by software engineers is contained in the software design description. The main purpose of this phase of waterfall model is to translate software design into source code.

After coding is completed each module is unit tested. Unit testing involves testing of each module in isolation from other modules. The purpose of unit testing is to determine correct working of the individual module.

### Integration and System Testing -

During this phase units of programs are

integrated and tested as a complete system to ensure that software requirement have been made. After system testing software is delivered to the customer.

## Implementation and Maintenance -

This is the process in which the developed system is handed over to the client.

Software maintenance is a task that every developer group has to face when the software is delivered to the customer site installed is operational software maintenance very broad activity that include error correction, enhancement, requirement deletion capability.

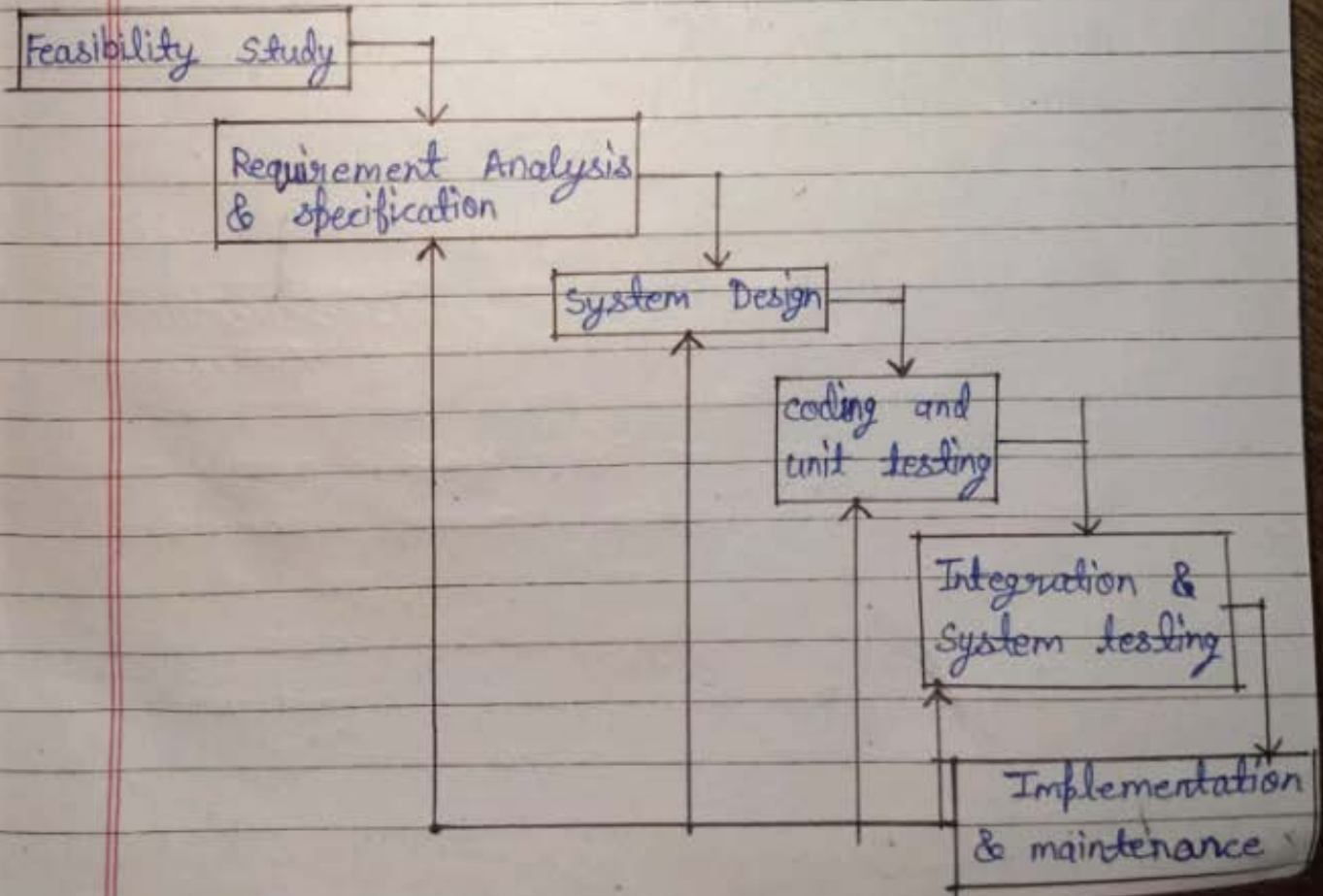
## Advantages of waterfall model -

- i) Waterfall model is an oldest and most widely used paradigm for software engineering.
- ii) This model provides a systematic and sequential approach of software development.

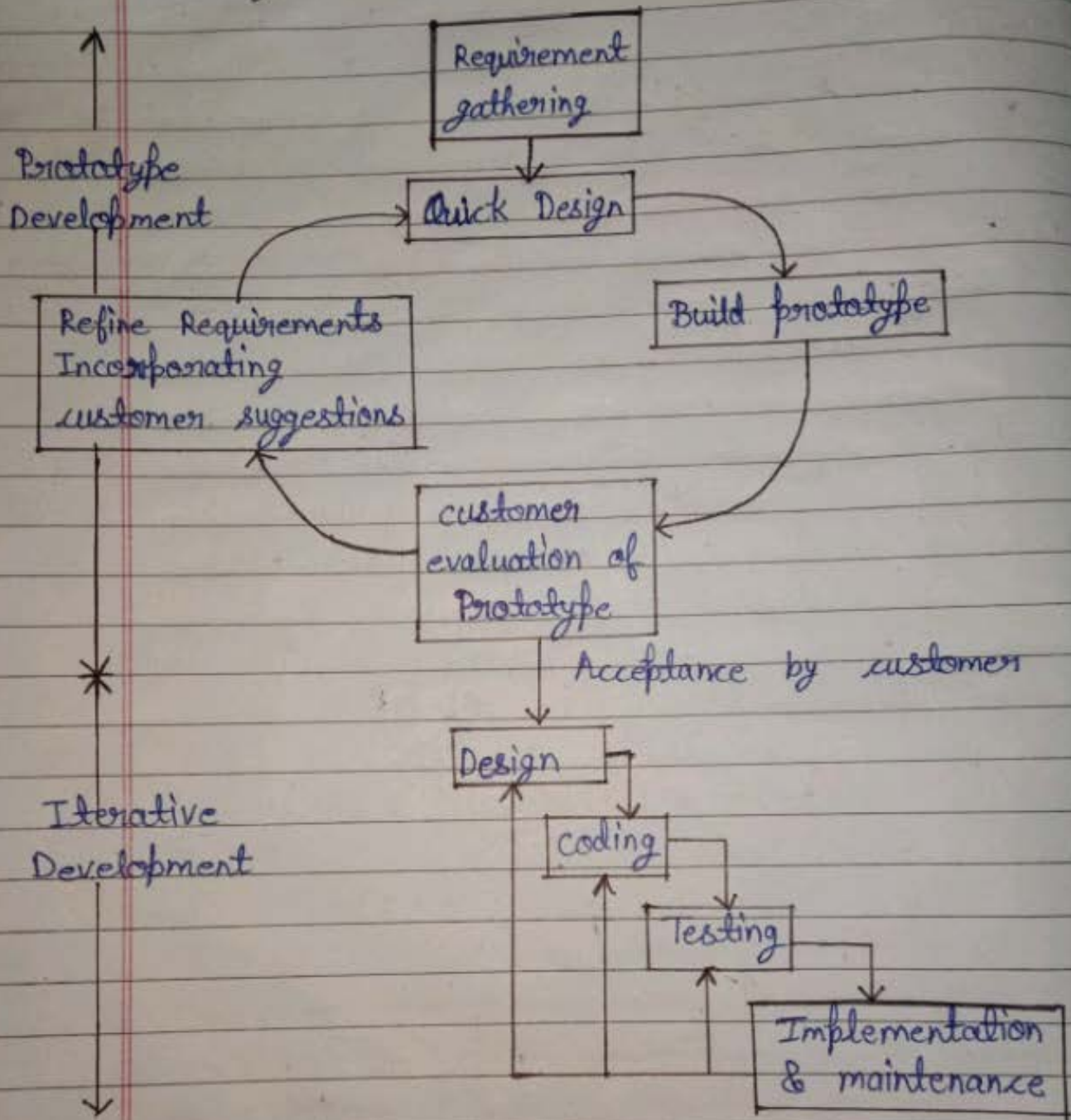
## Disadvantages of waterfall Model-

- i) This process model insist on a complete set of user requirement before commencement of design.
- ii) It is often difficult to customer to estate all the requirement activity explicitly.
- iii) Real project rarely follow the sequential flow that the model processes. It does not incorporate anykind of risk assessment.

## Iterative Waterfall Model -



### Prototype Model -



The prototype model suggest suggest that before development of actual software, a working prototype of the system should be build first. A prototype is a toy implementation of a system. A prototype means sample prototyping is a technique that

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allow for a reduce functionality or limited performance version of the actual software deliver to the user in the early stages of the development process. A ~~proton~~ prototype is usually built using several shortcuts. The shortcut might involve using inefficient, inaccurate or dummy functions. The development of the prototype itself involves design and coding. But this is not done in a formal manner.

### Advantages of Prototype Model-

- 1) It provides better understanding of the customers need.
- 2) It helps to critically examine the technical issues associated with the product development.
- 3) It is useful in the development of very large system.

### Disadvantages of Prototype Model-

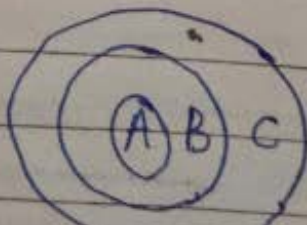
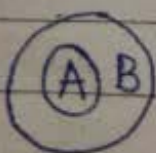
- 1) ~~Use~~ Use of inefficient, inaccurate or dummy functions may produce the desired result instead of performing the actual processing.

2) Prototype has to be discarded because of higher cost.

### Evolutionary Model -

This life cycle model is also referred to as the successive versions models and sometimes as the incremental model. In this life cycle model first a simple working system is build which sub-sequentially undergoes many functionality improvement, and additions until the designed system is successful & fully completed. The Evolutionary soft. development process is therefore sometimes referred as design a little, build a little, test a little, deploy a little model.

**Life Cycle Activities -** In the evolutionary life cycle model the soft. requirements are first broken down into several modules that can be incrementally constructed and delivered. The development team first develops the core module of the system.



A, B, C are modules of soft products that are incrementally developed and delivered.

### Advantages -

- 1) In this model the user gets a chance to experiment with a partially developed soft. much before the complete version of the system is released.
- 2) It is a very natural model to use in object orientated software developed project.

### Disadvantage -

- 1) The most popular problem it is difficult to divide the problem into several version that would be acceptable to the customer.

Spiral Model - This model was presented to combine the strength of various other models it incorporates the elements of the prototype along with the waterfall model. The diagrammatic representation of this model appears like a spiral with many loops the exact no. of loops of the spiral is not